



ANGULAR

COMPONENTS



codewithrutu_09

Components are the main building blocks for angular applications.

The Component contains data & user view interaction logic that defines how the view looks and behaves. A **View** in Angular refers to a template (HTML).

The Angular components are plain Javascript classes and defined using **@Component Decorator**. This Decorator provides the component with the view to display & Metadata about the component.

Creating a new component using the Angular CLI

1. From a terminal window, navigate to the directory containing your application.

2. Run the following command in terminal.



```
ng generate componnet <component-name>
```

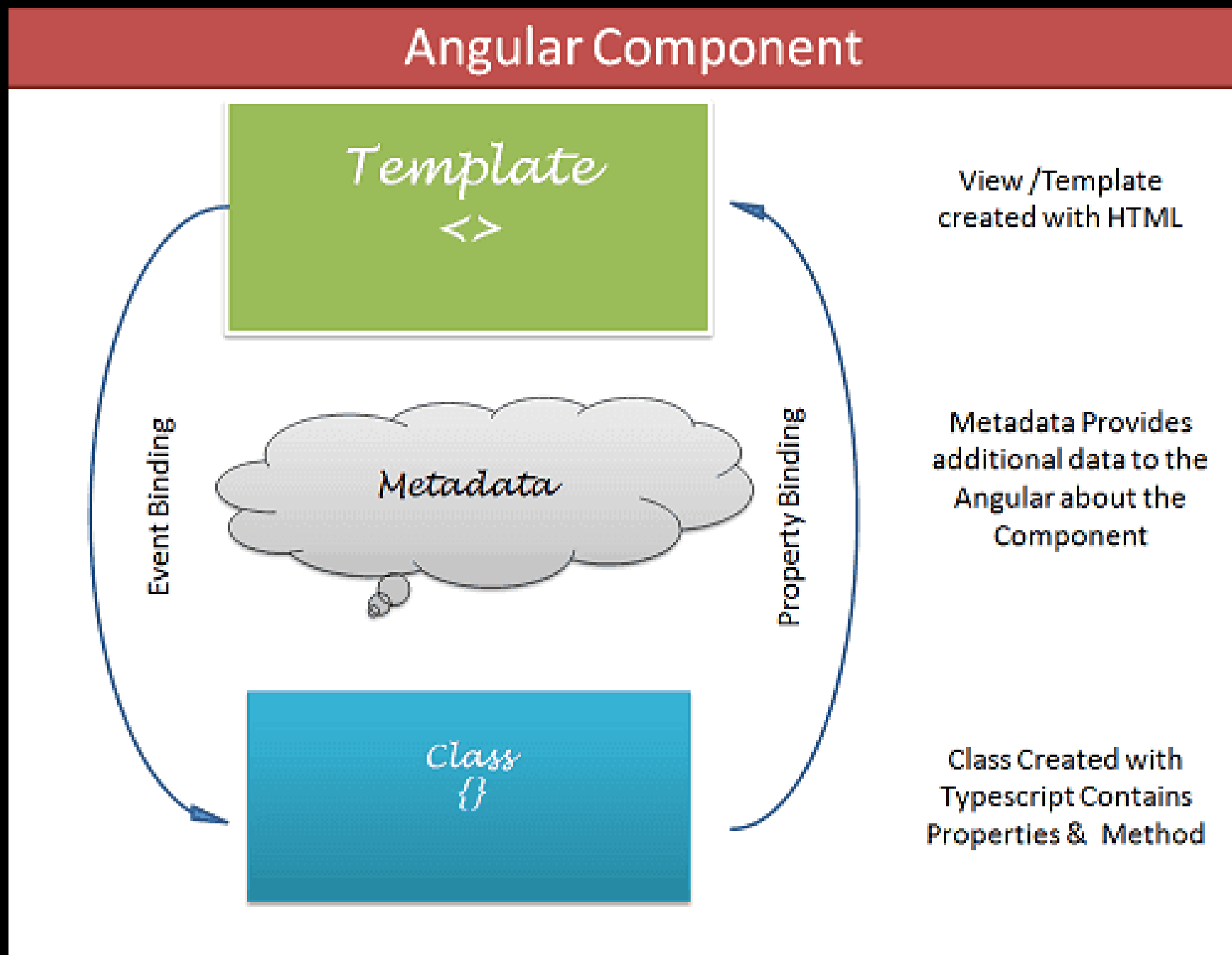
where the **<component-name>** is the name of your new component

By default, this command creates the following files:

- **A folder named after the component**
- **A Component ts file**
- **A Component template file**
- **A Component css file**
- **A Component testing specification file**

Each Angular App consists of at least one component which is known as **Root Component (app-root.component.html)**

The components consists of three main building blocks: Templates, Class & Metadata.



TEMPLATE (VIEW)

- The template defines the layout and content of the view.
- You can add **Angular directives, Angular Pipes & other Angular Components** on the template.
- The data to Template comes from the Component, which in turn gets it from a Angular Service. Using the data binding techniques, we can keep the Template in sync with the Component The templates can use the Event Binding or two way data binding to notify the component, when user changes something on the view.

CLASS

- The Class provides the data & logic to the View. It contains the Javascript code associated with Template (view).
- Class contains the Properties & Methods. The Properties of a class can be bind to the view using Data Binding.
- The simple Angular Class.



```
export class AppComponent
{
  title : string = "app"
}
```


METADATA

- Metadata provides **additional information about the component** to the Angular. Angular uses this information to process the class. We use the `@component` decorator to provide the Metadata to the Component.

@COMPONENT DECORATOR

- A decorator is a function that adds metadata to class, its methods & to its properties. The Components are defined with a `@component` class decorator.

- When Angular sees a class with `@component` decorator, it treats the **class as component**.
- A Decorator is **prefixed with @**.
- We can also build own decorators.

IMPORTANT COMPONENT METADATA PROPERTIES :

Selector:

Selector specifies the simple css selector. The Angular looks for the CSS selector in the template and renders the component there.

template/templateUrl:

The HTML template that **defines our view**. It tells Angular how to render the Component's view. The templates can be inline (using a template) or we can use an external template (using a templateUrl). The component can have **only one template**. You can either use inline template or external template and not both

styles/styleUrls:

The CSS styles or style sheets, that this component needs. Here we can use either **external stylesheet** (using **styleUrls**) or **inline styles** (using **style**).



codewithrutu_09

Was it Useful?

**Let me know your thoughts on this
component in the comments**